

Semantic Layer capture

Purpose	Best Format	Why
Human understanding	<code>.md</code>	Easy for developers, analysts, auditors
Semantic definitions	<code>.yaml</code>	Readable, structured, versionable
Strict validation	<code>.jsonschema</code> / <code>.py</code> tests	Prevents drift and ambiguity
App/API use	<code>.json</code> generated from YAML	Easy for services/tools to consume
Knowledge graph	RDF/OWL later, optional	Useful only if you need formal ontology reasoning

```
semantic-layer/  
├── README.md  
├── glossary.md  
├── entities.yaml  
├── relationships.yaml  
├── status-taxonomy.yaml  
├── business-rules.yaml  
├── metrics.yaml  
├── access-semantics.yaml  
├── ai-context-policy.yaml  
├── schemas/  
│   └── semantic-layer.schema.json  
├── tests/  
│   └── test_semantic_layer.py
```

Why YAML is best as the core

YAML is best because it is:

- readable by humans
- structured enough for machines
- easy for LLMs to generate and consume
- Git-friendly
- usable across models
- convertible to JSON
- good for entities, mappings, rules, statuses, policies, metrics

When YAML Is Not Enough

YAML alone is weak for:

- long explanations
- diagrams
- examples
- design rationale
- strict validation
- complex graph reasoning

.md	for explanation
.yaml	for source of truth
jsonschema	for structure validation
tests	for behaviour consistency

Markdown explains it.
YAML defines it.
JSON Schema validates it.
Tests protect it.
Generated JSON serves it.